1. Core Idea

Heart beats → blood pushed to face → skin colour flickers ∼1% → camera reads flicker → AI turns it into HR + BP. No cuff, no watch.

One-line summary

Patient looks at webcam 45s. Browser crops tiny 72 × 72 face frames, sends to Python server. 3 rPPG neural nets extract pulse wave, blend by quality, feed into a ResNet (trained on ICU data) → BP.

2. rPPG Basics

PPG = photoplethysmography = measure blood volume using light (finger clip / smartwatch green light).
rPPG = **remote** PPG = same idea, no contact, ordinary camera from a distance.
BVP = Blood Volume Pulse = the wavy pulse line over time.
SQI / SNR = signal quality = how clean the wave is vs noise. Higher = better.

Pulse signal is only ∼1% of skin colour. Noise is much bigger: head motion, lighting changes, webcam auto-brightness, video compression. Whole challenge = pull a 1% signal out of noisy video.

3. Two Systems (know both)

Project has 2 full vitals systems. Started with A, upgraded to B.

System	How it works
A (legacy, fallback)	POS maths algorithm → pulse wave → 15 features → 6 XGBoost models → BP. Endpoint `/predict-rgb`. Browser sends avg colour numbers. Trained on PPG-BP (219 people).
B (current, production)	3 deep rPPG nets → pulse waves → blend by quality → 1D-ResNet → BP. Endpoint `/predict-faces`. Browser sends 72 × 72 faces. Trained on RLAP + MIMIC-III.

Why upgrade A→B: POS is a fixed formula, can't learn; nets learned from thousands of videos. A used 3 small skin patches; B uses whole face. XGBoost trained on 219 people; ResNet on huge ICU data. A = 1 algorithm (fails together); B = 3 models (fault tolerant).

4. Full Pipeline (System B)

Browser: camera 30fps × 45s → MediaPipe finds 468 face points → quality check → crop face +25% pad, shrink to 72 × 72 → store raw RGB (∼1350 frames) → gzip + base64.
↓ HTTPS POST
Server: decode → average frame pairs (30→15 fps) → run 3 rPPG nets → blend 3 waves by SQI → HR+HRV from best model → ResNet on wave (bandpass → sliding windows → per-window BP) → blend with NHANES → classify (AHA 2017).
↓
Out: HR · BP (with 90% range) · HRV · category · confidence.

5. Step 1 — Browser Capture

File: `useVitalsCapture.ts`. All on patient's device.

Setting	Value & why
Frame rate	30 fps. Standard webcam; plenty of headroom for ∼180 bpm.
Duration	45s (↑ from 30). ∼1350 frames = cleaner signal.
Crop size	72 × 72 (↑ from 36 × 36, 4× detail). Enough for texture, tiny upload.
Steady wait	3s well-placed before recording starts.
Face-loss	pause after 2s if face gone.
Min frames	300 (= 30 × 10s), else reject with message.

Per frame: draw to hidden canvas → MediaPipe FaceLandmarker (468 points) → quality check → bounding box + 25% pad (edges carry signal) → resize 72 × 72, keep R,G,B (72 × 72 × 3 = 15552 bytes/frame).
Why crop: privacy (no background), small size, matches training format, stable to small head moves.

6. Step 2 — Transport

∼1350 frames × 15552 B ≈ 21 MB raw. Two steps shrink it:
Gzip (`CompressionStream`): neighbouring frames near-identical → compresses hugely.
Base64: binary → text so it fits in JSON (+33% size, firewall-friendly).

Key detail: real fps

Browser sends the **actual measured fps** (frames ÷ seconds), not 30. Webcams drift (e.g. 26.4). HR is computed from timing, so wrong fps = wrong HR. One line protects accuracy.

7. Step 3 — Frame Averaging

Server averages consecutive frames in pairs: (1,2)→1, (3,4)→1. Halves fps: 30→15 (`effective_fps`).

Sensor noise is random → averaging cancels it. Pulse is nearly same in both frames → survives. 15 fps still captures up to 450 bpm (Nyquist: 2 samples/cycle), far above any human HR. So: free noise cut, lose nothing. `effective_fps` used for all later timing.

8. Step 4 — The 3 rPPG Nets ⋆

Run 3 deep models on the same frames (via `open`).

Model	What it is
PhysNet	3D-CNN. Looks at a stack of frames together → sees change over time = the pulse. Classic, proven.
EfficientPhys	Fast light CNN. Works on differences between frames → removes static stuff (background, skin tone), keeps blood flow. Good on CPU.
FacePhys	Face-tuned net. Learns which face regions carry strongest signal (forehead, cheeks). Usually most accurate on clean video.

.rlap suffix = trained on the **RLAP** dataset (face videos + real contact-sensor ground truth).
Each returns: **BVP** (pulse wave), **HR** (bpm), **SQI** (self-trust), **HRV** (SDNN, RMSSD, pNN50, LF/HF, breathing).

Why 3 models, not 1

- **Accuracy:** diff architectures fail differently → blending cancels errors.
- **Fault tolerance:** each in own try/except; if one crashes/NaN, others continue.
- **Quality-aware:** each gives own SQI → trust the good one more.
- **Explainable:** all HR+SQI returned in diagnostics.

If **all 3** fail → HTTP 500, never invents a number.

9. Step 5 — Fusing the Waves

Combine 3 waves, but **not** plain average — weight by quality.
$w_i = \mathrm{SQI}_i / \sum \mathrm{SQI}$, $\mathrm{BVP} = \sum w_i \cdot \mathrm{BVP}_i$

Model	SQI	Weight
FacePhys	0.80	0.80/1.60 = 50%
PhysNet	0.50	0.50/1.60 = 31%
EfficientPhys	0.30	0.30/1.60 = 19%

Clearest signal counts most. Floor `max(sqi,0.01)` stops zero/negative weight. If total quality ≈0 → plain median. Waves trimmed to shortest length so they line up.

10. Step 6 — HR & HRV

HR number = from the **single best-SQI model**, NOT the blend.
best = arg max(SQI). Overall confidence = mean SQI.

Wave: blend all 3 (blending waves cuts noise).
HR: take best model only. Averaging 68, 72, noisy 95 would drag the answer to the bad one.

HRV = tiny gaps between beats; some variation = healthy. Metrics: SDNN (overall spread), RMSSD (short-term, rest&digest), pNN50 (% gaps >50ms), LF/HF (stress balance), breathing rate.
Quality gate: if mean SQI < 0.1 → HTTP 422 "too low, better lighting". Never guesses.

11. Step 7 — ResNet BP ⋆

Clean wave → BP via **1D-ResNet**, pretrained on **MIMIC-III** ICU data. File `lab_bp_model.py`, weights ∼23 MB.
Why a wave gives BP: pulse **shape** carries it. Stiff arteries / high pressure → sharper rise, notch (reflected-wave bump) sits differently. Net learns these from raw wave.
Stage 1 — clean
Detrend (remove drift) → adaptive bandpass → scale 0..1.
Adaptive bandpass: we already know HR from step 6, so narrow the filter tight around it instead of fixed 0.7–3 Hz.
f_{hr} = HR/60; low= $\max(0.5, f_{hr} - 0.5)$; high= $\min(f_s/2 - 0.1, f_{hr} + 0.5)$. e.g. 72 bpm → keep 0.7–1.7 Hz only → throws away far more noise. `filtfilt` = forward+backward so no time shift.
Stage 2 — sliding windows
Net needs exactly **437 samples**. Window of 437 slides by **80** each step → heavy overlap → each part seen ∼5× → dozens of BP estimates. Each window → [SBP, DBP].
Stage 3 — combine windows
Each window scored by quality (Welch PSD: % power in pulse band). Quality-weighted average → clean windows count more. If all noisy → median fallback. Range from spread: $q05$ & $q95$ of window preds → 90% interval.

Honest uncertainty

Interval is NOT guessed. Windows agree → narrow range = confident. Windows disagree → wide range = doctor sees it's uncertain.

12. Step 8 — NHANES Blend

Final safety net. Weak signal → don't trust model 100%. Mix with what's statistically normal for that age+sex.
NHANES = CDC national health survey → average BP by age & sex (e.g. 45yo male ≈ 124.7/77.2). Ages between = linear interpolation. Hard-coded lookup table, not trained.
BMI shift: BMI = $w/(h/100)^2$. If >25 raise expected BP; if <18.5 lower it.
Blend: $b = \min(0.85, \max(0, \mathrm{SQI}))$

SBP $= b \cdot \text{SBP}_{res} + (1-b) \cdot \text{SBP}_{nh}$; same for DBP. $b < 0.5$ → flagged `signal_source: fallback`.

Quality	Blend	Answer based on
≥0.85	0.85	85% camera, 15% pop
0.60	0.60	60% camera, 40% pop
0.30	0.30	30% camera, 70% pop (fallback)

Even a perfect signal never gets 100% trust — camera BP is an estimate, so population prior keeps ≥15% pull. Drags wild predictions back to reality. Raw, NHANES, and blend values all returned → fully transparent.

13. Step 9 — Classify (AHA 2017)

Checked most-severe first, uses OR (either number high raises category):

Category	Rule
Hypertensive Crisis	SBP>180 or DBP>120
Stage 2	SBP≥140 or DBP≥90
Stage 1	SBP≥130 or DBP≥80
Elevated	SBP≥120 and DBP<80
Normal	otherwise

HR: <60 Bradycardia · 60–100 Normal · >100 Tachycardia.

14. Datasets ⋆ (most-asked)

Dataset	What / used for
RLAP	Face videos + contact-sensor pulse truth. Trained the 3 rPPG nets (`.rlap`). Production.
MIMIC-III	Huge ICU database: PPG + arterial BP waveforms (arterial line = beat-by-beat truth). Pretrained the 1D-ResNet. Production.
PPG-BP	219 subjects, 657 finger recordings @1000Hz, cuff BP labels (Figshare). Trained 6 XGBoost models. Legacy.
NHANES	CDC survey: avg BP by age/sex. Not trained — statistical prior table. Production (blending).

ResNet learned from **contact ICU** waves but runs on **camera** waves from healthy people at home. Gap is real → that's exactly why output is blended with NHANES and capped at 85% trust. Mitigated, not eliminated.

15. Legacy System A (compact)

POS (Plane-Orthogonal-to-Skin, Wang 2017): pure maths, no learning. Blood-colour change lies in one RGB direction, noise in another → project along right plane → pulse survives, noise cancels.
Faces: 3 skin patches (forehead + 2 cheeks) via landmarks; YCrCb skin mask drops hair/shadow. Flat, well-lit, vessel-rich spots.
Chain: POS per patch → SNR-weight to 1 wave → **resample 30→1000 Hz** → 15 features → 6 XGBoost (quantile) → blend + classify.

15 features (System A only)

HR (3): hr_fft, hr_peaks, hr_final.
HRV (3): rmssd, sdnn, pnn50.
Shape/morphology (6, the real BP signal): pulse_width, rise_time, fall_time, peak_amplitude, notch_ratio, aug_index.
Quality (3): signal_quality, peak_count, signal_std.
ResNet (System B) skips these — learns its own from the raw wave.

Quantile regression: predict a percentile not the mean. Train q05/q50/q95 → q50 = answer, q05–q95 = 90% interval. 2 targets × 3 quantiles = 6 models. **Coverage** metric: true value should land inside the range ∼90% of the time.

16. Quality Gates

System refuses bad readings. Order: face present/steady 3s → face-loss >2s pauses → ≥300 frames → decode ok → validation (frames/crop/fps/age) → each model valid HR+SQI → ≥1 model works (else 500) → mean SQI ≥0.1 (else 422) → ≥437 samples for ResNet → ResNet ok.

Graceful degradation

If BP fails, request does NOT crash — patient still gets HR + HRV, response says why BP is missing. Partial results beat none, as long as nothing is faked.

17. Privacy

Never sent: video, audio, background, others in frame, full-res face.
Sent: gzipped 72 × 72 face thumbnails + age/sex/height/weight only.
MediaPipe runs in-browser. Server keeps frames in memory a few seconds, never writes to disk. DB stores only final numbers.

18. Specs Cheat-Sheet

Item	Value
Scan / rate	45s / 30 fps (real fps sent)
Frames	∼1350
Crop	$72 \times 72 \times 3 = 15552$ B/frame
Padding / landmarks	25% / 468 (MediaPipe)
Averaging	pairs → 15 fps effective
Ensemble	3 nets (.rlap)
BP model	1D-ResNet, 12 blocks, ∼23 MB
ResNet window	437 samples, step 80
Max camera blend	0.85
Min SQI	0.10
Legacy models	6 XGBoost quantile
HR accuracy	±5–8 bpm (good light)
Interval	90% (q05–q95)

19. Limitations (say honestly)

- Not a medical device — estimate, supports a doctor, never replaces a cuff.
- Domain gap (ICU contact → camera); mitigated by NHANES.
- Lighting & motion sensitive; dim light → pushes to population avg.
- Skin tone / makeup / facial hair reduce usable skin.
- Camera BP still an active research area; HR is well-established.
- Small legacy set (219). Needs Python service reachable.

20. ResNet Architecture

Input $(1, 1, 437)$ → Conv1d(1→64, k7, s2) + BN + ReLU + MaxPool → 4 stages of 3 residual blocks each, widening 64→128→256→512 (stride 2 halves length, doubles depth) → AdaptiveAvgPool1d(1) → Linear(512→2) → [SBP, DBP].
ResNet = skip connection adds block input to its output → block learns only the *change* (residual) → gradients flow clean through deep nets.
1D = input is a time signal not an image → Conv1d slides along time only, learning shapes (sharp rise, wide plateau, second bump).

All Formulae

Name	Formula
Fusion weight	$w_i = \text{SQI}_i / \sum \text{SQI}$
Fused wave	$\text{BVP} = \sum w_i \, \text{BVP}_i$
Best model (HR)	$\arg\max(\text{SQI})$
Eff. fps	fps/2
Nyquist limit	15 fps → 450 bpm max
Adaptive band	$f_{hr} = \text{HR}/60$; low= $\max(0.5, f_{hr} - 0.5)$; high= $\min(f_s/2 - 0.1, f_{hr} + 0.5)$
BMI	$w/(h/100)^2$
Blend factor	$b = \min(0.85, \max(0, \text{SQI}))$
Final BP	$b \cdot \text{res} + (1-b) \cdot \text{nhanes}$

PPG vs rPPG: PPG = contact light sensor. rPPG = remote, camera, no contact.
Wave fusion vs HR pick: wave = blend all 3. HR = best model only.
System A vs B: A = POS + XGBoost (legacy, colour numbers). B = 3 nets + ResNet (live, face images).
RLAP vs MIMIC-III: RLAP trains the rPPG nets (face→wave). MIMIC-III trains the ResNet (wave→BP).
PPG-BP vs NHANES: PPG-BP = training data (XGBoost). NHANES = prior table, not trained.
2D vs 1D ResNet: 2D = images (H×W). 1D = signal, Conv slides along time only.
SQI vs coverage: SQI = signal cleanliness. Coverage = % of true values inside the 90% range.
Fixed vs adaptive bandpass: fixed = 0.7–3 Hz always. Adaptive = narrow around detected HR.

Exam Quick Hits

- Pulse = ∼1% skin colour flicker; whole job = pull it from noise.
- 3 nets = accuracy + fault tolerance + quality-aware + explainable.
- Blend waves by SQI; take HR from best model.
- ResNet shape → BP; adaptive bandpass around known HR; 437-sample windows, step 80.
- 90% range = spread of window preds (honest uncertainty).
- Cap camera trust at 0.85 → NHANES prior always ≥15%.
- 4 datasets: RLAP, MIMIC-III, PPG-BP, NHANES.
- AHA 2017 rules, OR logic, most-severe-first.
- Privacy: only 72 × 72 crops leave device; nothing to disk.
- Never fakes a number — rejects or degrades gracefully.